Problem 1. Several stones are placed on an infinite (in both directions) strip of squares. As long as there are at least two stones on a single square, you may pick up two such stones, then move one to the preceding square and one to the following square. Is it possible to return to the starting configuration after a finite sequence of such moves?

Ans. Label the strip with consecutive integers, and let n_i denote the label of the square containing stone #1. Let $X = \sum_i n_i^2$, and consider what happens to X each time we do a move. First X decreases by $2t^2$ as we remove two stones from some square t. Next, X increases by $(t-1)^2+(t+1)^2 = 2t^2+2$ as we replace the stones in squares $t-1$ and $t+1$. Therefore, every move causes X to increase by exactly 2. In particular, after any sequence of moves, X will always be higher than where it began, so we could not possibly be in the same position we began in.

Problem 2. Numbers $\frac{49}{1}, \frac{49}{2}, ..., \frac{49}{97}$ are writen on a blackboard. Each time, we can replace two numbers (like a, b) with $2ab - a - b + 1$. After 96 times doing that prenominate action, one number will be left on the board. Find all the possible values fot that number.

Ans. We observe that $2(2ab-a-b+1)-1 = (2a-1)(2b-1)$. Then there is an invariant $\Pi = \Pi_n = \prod_{i=1}^{n}(2a_i-1)$, where n is the number of the numbers written on the blackboard! Since $\Pi_{97} = 1$, it follows the last number will also be $\Pi_1 = 1$.

Problem 3. A circle is divided into six sectors. Then the numbers $1, 0, 1, 0, 0, 0$ are written into the sectors. You may increase two neighboring numbers by 1. Is it possible to equalize all numbers by a sequence of such steps?

Ans. Let the numbers in the sectors be a, b, c, d, e, f. Initially, $(a, b, c, d, e, f) : (1, 0, 1, 0, 0, 0)$.

Consider the sum $S = a - b + c - d + e - f$. If two neighboring numbers increase by 1, S remains constant, hence S is invariant.

Notice that if all numbers are equal, then $S = 0$. Yet, initially $S = 2$. Hence it is not possible.

Problem 4. Given 2004 vertices $a_1, a_2, \cdots, a_{2004}$ circling a table in clockwise order. Initially, a_1 is labeled 0 and $a_2, a_3, \cdots, a_{2004}$ are labeled 1. In every step, we choose three vertices a_{i-1}, a_i, a_{i+1} with labels a, b, c and change them into $1 - a, 1 - b, 1 - c$ respectively. Is it possible to change all labels into 0?

Ans. The answer is no; between any two numbers write down their absolute difference, and consider the circle containing only these absolute differences. To be more rigorous, let $b_i = |a_{i+1} - a_i|$, where we take $a_{2005} = a_1$. Then $b_1 = b_2 = 1$ and $b_3 = b_4 = \cdots = b_{2004} = 0$. Observe that an operation consists of choosing two numbers and flipping them. To be more rigorous, flipping vertices a_{i-1}, a_i, a_{i+1} corresponds to flipping the numbers b_{i-1} and b_{i+2}. We show that it's impossible to have $b_i = 0$ for all $1 \le i \le 2004$; this clearly implies the condition.

Consider the numbers $b_1, b_4, b_7, \cdots, b_{2002}$, i.e. those with indices which are 1 mod 3. There are $\dfrac{2004}{3} = 668$ of these, and initially only one of these is labeled 1. Any operation preserves the parity of the number of 1s; that is, it is always odd no matter how many times we operate. Thus we can never have $b_1 = b_4 = \cdots = 0$, which clearly implies the result.

Problem 5. (USAMO 1994) The sides of a 99-gon are initially colored so that consecutive sides are red, blue, red, blue, . . . , red, blue, yellow. We make a sequence of modifications in the coloring, changing the color of one side at a time to one of the three given colors (red, blue, yellow), under the constraint that no two adjacent sides may be the same color. By making a sequence of such modifications, is it possible to arrive at the coloring in which consecutive sides are red, blue, red, blue, red, blue, . . . , red, yellow, blue?

Ans. We will construct the invariant as follows: for the i^{th} side $(1 \le i \le 99)$, we label $a_i = 0$ if the side is red, $a_i = 1$ if the side is blue, and $a_i = 2$ if the side is yellow. We can easily see that

$$P = (a_2 - a_1)(a_3 - a_2) \cdots (a_{99} - a_{98})(a_1 - a_{99}) \pmod 3$$

does not change since all modifications turn three consecutive sides of colors x, y, x into x, z, x where $x \neq y, z$, but

$$(y - x)(x - y) \equiv (z - x)(x - z) \equiv -1 \pmod 3$$

since the only nonzero quadratic residue modulo 3 is 1. But now we see that the original P is (starting with red and ending with yellow, $a_{2k-1} = 0$ and $a_{2k} = 1$, except for $a_{99} = 2$),

$$[(1)(-1)\cdots(1)(-1)(1)](1)(-2) \equiv (-1)^{\frac{97-1}{2}}(-2) \equiv 1 \pmod 3$$

while the new P is (starting with blue and ending with yellow, $a_{2k-1} = 1$ and $a_{2k} = 0$, except for $a_{99} = 2$),

$$[(-1)(1)\cdots(-1)(1)(-1)](-1)(2) \equiv (-1)^{\frac{97+1}{2}}(-2) \equiv 2 \pmod 3,$$

which is a contradiction.

Problem 6. (Romanian TST 2002) After elections, every parliament member (PM), has his own absolute rating. When the parliament set up, he enters in a group and gets a relative rating. The relative rating is the ratio of its own absolute rating to the sum of all absolute ratings of the PMs in the group. A PM can move from one group to another only if in his new group his relative rating is greater. In a given day, only one PM can change the group. Show that only a finite number of group moves is possible.

Ans. This is clearly a problem that asks for finding the right (semi)invariant!

Denote the absolute rating of a PM x by $r(x) > 0$. Say there are $n > 1$ groups $(G_k)_{1\leq k\leq n}$, all non-empty. Count the moves by $m = 1, 2, \ldots$. Then $s_0 = \sum_{k=1}^{n}\left(\sum_{x\in G_k} r(x)\right)$ is an invariant - the sum of all absolute ratings before any move was made, but the same after any number m of moves, so $s_m = s_0$, constant. This is of no use.

Let us rather consider the semi-invariant $\sigma_m = \sum_{k=1}^{n}\left(\sum_{x\in G_k} r(x)\right)^2$. Let us say at the next move a PM x moves from a group G to a group H. Denote by $a = \sum_{y\in G\setminus\{x\}} r(y)$ and by $b = \sum_{z\in H} r(z)$. Then it means the relative rating of x increased, i.e. $\dfrac{r(x)}{a + r(x)} < \dfrac{r(x)}{b + r(x)}$, or $a > b$.

Now it is easy to compute that $\sigma_{m+1} - \sigma_m = (b + r(x))^2 + a^2 - b^2 - (a + r(x))^2 = 2r(x)(b - a) < 0$, so $(\sigma_m)_{m\geq 0}$ is a decreasing sequence, lower bounded by 0. But the value $2r(x)(a - b)$, by which it decreases at a move, cannot be too small!

The possible sums of absolute ratings of a bunch of PM's can only take a finite number of values - for the 2 power the total number of PM's of all possibilities (less 1, for the empty set). Therefore $2r(x)(a - b) \geq \varepsilon > 0$ for some ε. By the principle of infinite descent, the number of possible moves is therefore finite, since σ_m decreases by at least ε at each move, while having to remain a positive number at all times.

Problem 7. (2022 China TST, Test 1, P3) Let a, b, c, p, q, r be positive integers with $p, q, r \geq 2$. Denote

$$Q = \{(x, y, z) \in \mathbb{Z}^3 : 0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c\}.$$

Initially, some pieces are put on the each point in Q, with a total of M pieces. Then, one can perform the following three types of operations repeatedly: (1) Remove p pieces on (x, y, z) and place a piece on $(x-1, y, z)$; (2) Remove q pieces on (x, y, z) and place a piece on $(x, y-1, z)$; (3) Remove r pieces on (x, y, z) and place a piece on $(x, y, z-1)$.

Find the smallest positive integer M such that one can always perform a sequence of operations, making a piece placed on $(0, 0, 0)$, no matter how the pieces are distributed initially.

Ans. The answer is $p^a q^b r^c$. Let $f(x, y, z)$ be the number of pieces on the lattice point (x, y, z). For sets X, Y, Z, let $X \times Y \times Z = \{(m, n, t) | m \in X, n \in Y, t \in Z\}$

I first show $M \geq p^a q^b r^c$. Let

$$N = \sum_{0\leq x\leq a, 0\leq y\leq b, 0\leq z\leq c} f(x, y, z) p^{-x} q^{-y} r^{-z}$$

Note N is an invariant. In the end, $f(0, 0, 0) \geq 1$. Therefore, if $M < p^a q^b r^c$, setting $f(a, b, c) = M$ gives $N < 1$, contradiction.

Now I show $p^a q^b r^c$ always works. We proceed by strong induction, and we will use the inductive hypothesis very strongly. The key intuition is that almost all the pieces must be on (a, b, c) while there must be a piece on $x = 0$. Combining these two gives a way to place a cell in $(0, 0, 0)$.

Case 1: there are no pieces with $x = 0$. We group the $p^a q^b r^c$ pieces into p random groups of $p^{a-1} q^b r^c$ pieces. Each move uses pieces in the same group only. By inductive hypothesis on $(a-1, b, c)$ on $\{1, \cdots, a\} \times \{0, \cdots, b\} \times$

$\{0, \cdots, c\}$, each group of $p^{a-1}q^br^c$ piece can result in one piece on $(1,0,0)$. Now there are p pieces on $(1,0,0)$, we can use these to get one piece on $(0,0,0)$.

Case 2: There is a piece with $x = 0$, a piece with $y = 0$ and a piece with $z = 0$.

Claim: There are at most $(b+1)(c+1) - 2$ pieces in $\{0, \cdots, a-1\} \times \{0, \cdots, b\} \times \{0, \cdots, c\}$.

Proof: We can move $\lfloor \frac{f(a,n,t)}{p} \rfloor$ pieces from (a, n, t) to $(a-1, n, t)$. After transferring all cells from (a, n, t) to $(a-1, n, t)$ for $0 \le n \le b, 0 \le t \le c$.

Let A be the number of pieces in $\{0, \cdots, a-1\} \times \{0, \cdots, b\} \times \{0, \cdots, c\}$. If $A + \sum_{0 \le n \le b, 0 \le t \le c} \lfloor \frac{f(a,n,t)}{p} \rfloor \ge p^{a-1}q^br^c$, by inductive hypothesis on $(a-1, b, c)$, we can place a cell on $(0,0,0)$. Therefore, it suffices to handle the case where $A + \sum_{0 \le n \le b, 0 \le t \le c} \lfloor \frac{f(a,n,t)}{p} \rfloor \le p^{a-1}q^br^c - 1$ (1)

We also have $A + \sum_{0 \le n \le b, 0 \le t \le c} f(a, n, t) = p^a q^b r^c$ (2) because there are a total of $p^aq^br^c$ pieces on the board.

$p(1) - (2)$ gives $(p-1)A - \sum_{0 \le n \le b, 0 \le t \le c} R(f(a, n, t), p) \le -p$

It follows that $(p-1)A - \sum_{0 \le n \le b, 0 \le t \le c} (p-1) \le -p$ so $A \le (b+1)(c+1) - 2$.

Using similar arguments, we can see that the number of stones in $\{0, \cdots, a\} \times \{1, \cdots, b\} \times \{0, \cdots, c\}$ is at most $(a+1)(c+1)-2$ and the number of stones in $\{0, \cdots, a\} \times \{0, \cdots, b\} \times \{1, \cdots, c\}$ is at most $(a+1)(b+1)-2$. This means the number of stones not in (a, b, c) is at most $(a+1)(b+1) + (b+1)(c+1) + (c+1)(a+1) - 6$, so the number of stones in (a, b, c) is at least $p^aq^br^c - (a+1)(b+1) + (b+1)(c+1) + (c+1)(a+1) + 6$.

Let $X = (0, k, l)$ be a square with at least one cell. If I can transfer $q^kr^l - 1$ stones to X, I win by inductive hypothesis on $a = 0, b = k, c = l$. To transfer $q^kr^l - 1$ stones to X, I need $(q^kr^l - 1)(p^aq^{b-k}r^{c-l}) = p^aq^br^c - p^aq^{b-k}r^{c-l} \le p^aq^br^c - p^a$. Therefore, if $p^a \ge (a+1)(b+1) + (b+1)(c+1) + (c+1)(a+1) - 6$ I am done. Similarly, if $q^b, r^c \ge (a+1)(b+1) + (b+1)(c+1) + (c+1)(a+1) - 6$ I am also done.

It remains to handle small cases?

Problem 8. (Putnam 2008 A3) Start with a finite sequence $a_1, a_2, \ldots, a_n$ of positive integers. If possible, choose two indices $j < k$ such that a_j does not divide a_k and replace a_j and a_k by $gcd(a_j, a_k)$ and $lcm(a_j, a_k)$, respectively. Prove that if this process is repeated, it must eventually stop and the final sequence does not depend on the choices made. (Note: *gcd* means greatest common divisor and *lcm* means least common multiple.)

Ans. We first prove that the process stops. Note first that the product $a_1 \cdots a_n$ remains constant, because $a_ja_k = gcd(a_j, a_k)lcm(a_j, a_k)$. Moreover, the last number in the sequence can never decrease, because it is always replaced by its least common multiple with another number. Since it is bounded above (by the product of all of the numbers), the last number must eventually reach its maximum value, after which it remains constant throughout. After this happens, the next-to-last number will never decrease, so it eventually becomes constant, and so on. After finitely many steps, all of the numbers will achieve their final values, so no more steps will be possible. This only happens when a_j divides a_k for all pairs $j < k$.

We next check that there is only one possible final sequence. For p a prime and m a nonnegative integer, we claim that the number of integers in the list divisible by p^m never changes. To see this, suppose we replace a_j, a_k by $gcd(a_j, a_k), lcm(a_j, a_k)$. If neither of a_j, a_k is divisible by p^m, then neither of $gcd(a_j, a_k), lcm(a_j, a_k)$ is either. If exactly one a_j, a_k is divisible by p^m, then $lcm(a_j, a_k)$ is divisible by p^m but $gcd(a_j, a_k)$ is not.

$gcd(a_j, a_k), lcm(a_j, a_k)$ are as well.

If we started out with exactly h numbers not divisible by p^m, then in the final sequence $a'_1, \ldots, a'_n$, the numbers $a'_{h+1}, \ldots, a'_n$ are divisible by p^m while the numbers $a'_1, \ldots, a'_h$ are not. Repeating this argument for each pair (p, m) such that p^m divides the initial product $a_1, \ldots, a_n$, we can determine the exact prime factorization of each of $a'_1, \ldots, a'_n$. This proves that the final sequence is unique.

Remark: (by David Savitt and Noam Elkies) Here are two other ways to prove the termination. One is to observe that $\prod_j a_j^j$ is *strictly* increasing at each step, and bounded above by $(a_1 \cdots a_n)^n$. The other is to notice that a_1 is nonincreasing but always positive, so eventually becomes constant; then a_2 is nonincreasing but always positive, and so on.

Reinterpretation: For each p, consider the sequence consisting of the exponents of p in the prime factorizations of $a_1, \ldots, a_n$. At each step, we pick two positions i and j such that the exponents of some prime p are in the wrong order at positions i and j. We then sort these two position into the correct order for every prime p simultaneously.

It is clear that this can only terminate with all sequences being sorted into the correct order. We must still check that the process terminates; however, since all but finitely many of the exponent sequences consist of all zeroes, and each step makes a nontrivial switch in at least one of the other exponent sequences, it is enough to check the case of a single exponent sequence. This can be done as in the first solution.

Remark: Abhinav Kumar suggests the following proof that the process always terminates in at most $\binom{n}{2}$ steps. (This is a variant of the worst-case analysis of the *bubble sort* algorithm.)

Consider the number of pairs (k, l) with $1 \leq k < l \leq n$ such that a_k does not divide a_l (call these *bad pairs*). At each step, we find one bad pair (i, j) and eliminate it, and we do not touch any pairs that do not involve either i or j. If $i < k < j$, then neither of the pairs (i, k) and (k, j) can become bad, because a_i is replaced by a divisor of itself, while a_j is replaced by a multiple of itself. If $k < i$, then (k, i) can only become a bad pair if a_k divided a_i but not a_j, in which case (k, j) stops being bad. Similarly, if $k > j$, then (i, k) and (j, k) either stay the same or switch status. Hence the number of bad pairs goes down by at least 1 each time; since it is at most $\binom{n}{2}$ to begin with, this is an upper bound for the number of steps.

Remark: This problem is closely related to the classification theorem for finite abelian groups. Namely, if $a_1, \ldots, a_n$ and $a'_1, \ldots, a'_n$ are the sequences obtained at two different steps in the process, then the abelian groups $/a_1 \times \cdots \times /a_n$ and $/a'_1 \times \cdots \times /a'_n$ are isomorphic. The final sequence gives a canonical presentation of this group; the terms of this sequence are called the *elementary divisors* or *invariant factors* of the group.

Remark: (by Tom Belulovich) A *lattice* is a partially ordered set L in which for any two $x, y \in L$, there is a unique minimal element z with $z \geq x$ and $z \geq y$, called the *join* and denoted $x \wedge y$, and there is a unique maximal element z with $z \leq x$ and $z \leq y$, called the *meet* and denoted $x \vee y$. In terms of a lattice L, one can pose the following generalization of the given problem. Start with $a_1, \ldots, a_n \in L$. If $i < j$ but $a_i \not\leq a_j$, it is permitted to replace a_i, a_j by $a_i \vee a_j, a_i \wedge a_j$, respectively. The same argument as above shows that this always terminates in at most $\binom{n}{2}$ steps. The question is, under what conditions on the lattice L is the final sequence uniquely determined by the initial sequence?

It turns out that this holds if and only if L is *distributive*, i.e., for any $x, y, z \in L$,

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z).$$

(This is equivalent to the same axiom with the operations interchanged.) For example, if L is a *Boolean algebra*, i.e., the set of subsets of a given set S under inclusion, then $\wedge$ is union, $\vee$ is intersection, and the distributive law holds. Conversely, any finite distributive lattice is contained in a Boolean algebra by a theorem of Birkhoff. The correspondence takes each $x \in L$ to the set of $y \in L$ such that $x \geq y$ and y cannot be written as a join of two elements of $L \setminus \{y\}$. (See for instance Birkhoff, *Lattice Theory*, Amer. Math. Soc., 1967.)

On one hand, if L is distributive, it can be shown that the j-th term of the final sequence is equal to the meet of $a_{i_1} \wedge \cdots \wedge a_{i_j}$ over all sequences $1 \leq i_1 < \cdots < i_j \leq n$. For instance, this can be checked by forming the smallest subset L' of L containing $a_1, \ldots, a_n$ and closed under meet and join, then embedding L' into a Boolean algebra using Birkhoff's theorem, then checking the claim for all Boolean algebras. It can also be checked directly (as suggested by Nghi Nguyen) by showing that for $j = 1, \ldots, n$, the meet of all joins of j-element subsets of $a_1, \ldots, a_n$ is invariant at each step.

On the other hand, a lattice fails to be distributive if and only if it contains five elements $a, b, c, 0, 1$ such that either the only relations among them are implied by

$$1 \geq a, b, c \geq 0$$

(this lattice is sometimes called the *diamond*), or the only relations among them are implied by

$$1 \geq a \geq b \geq 0, \qquad 1 \geq c \geq 0$$

(this lattice is sometimes called the *pentagon*). (For a proof, see the Birkhoff reference given above.) For each of these examples, the initial sequence a, b, c fails to determine the final sequence; for the diamond, we can end up with $0, *, 1$ for any of $* = a, b, c$, whereas for the pentagon we can end up with $0, *, 1$ for any of $* = a, b$.

Consequently, the final sequence is determined by the initial sequence if and only if L is distributive.

Problem 9. (Korea Final Round 2009) 2008 white stones and 1 black stone are in a row. An 'action' means the following: select one black stone and change the color of neighboring stone(s). Find all possible initial position of the black stone, to make all stones black by finite actions.

Ans. Let the stones be $a_1, a_2.....a_{2009}$. To each white stone with k black stones to its right assign the number $(-1)^k$. Let S be the sum of the assigned numbers. One can check that S is invariant, except when we act on a_1 or a_{2009}. In view of this we will add white stones a_0 and a_{2010} at the start and the end. Then S is invariant when we can only act on a_1 to a_{2009}. Now consider the ending configuration, i.e. a_1 to a_{2009} are all black, consider the 4 possibilities of what a_0 and a_{2010} can be (both stone can be either white or black), we see that S can only be 0 or ± 1. For the initial position, since we originally set a_0 and a_{2010} be white, we can easily see that for all initial configuration S congruents to 0 modulus 2. Hence the only possible initial configuration is when $S = 0$, which occurs only when the black stone is in the middle, i.e. the initial position is

$$\overbrace{www\dots www}^{1004}b\overbrace{www\dots www}^{1004}.$$